THE H^0 OF THE INTERNAL HOM INTO A K-INJECTIVE COMPLEX COMPUTES HOM IN THE DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the displayed chain for the global statement and the localisation step through the restriction lemma for K-injective complexes.

Editorial note. The label lemma-RHom-into-K-injective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The cohomology of the internal hom complex as homotopy classes of maps. Same chapter, the equation identifying the cohomology of the internal hom complex with the homotopy category Hom group. Supplies the first equality of the displayed chain.
- Localisation of K-injectivity to an open. Same chapter, the lemma stating that the restriction of a K-injective complex to a localisation is again K-injective. Supplies the second equality, since a K-injective representative computes Hom in the derived category.

Source reference. Original equation 0A5X: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8601–8601. Prerequisite pointer invoked at proof line(s) 8746; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0A8Z: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8608–8608. Prerequisite pointer invoked at proof line(s) 8750; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08BS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4314–4314. Prerequisite pointer invoked at proof line(s) 8748; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let L and M be objects of $D(\mathcal{O})$. Let $\mathcal{I}^\bullet$ be a K-injective complex of $\mathcal{O}$-modules representing M. Let $\mathcal{L}^\bullet$ be a complex of $\mathcal{O}$-modules representing L. Then*

$$H^0(\Gamma(U, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet))) = \mathrm{Hom}_{D(\mathcal{O}_U)}(L|_U, M|_U)$$

for all $U \in \mathrm{Ob}(\mathcal{C})$. Similarly, $H^0(\Gamma(\mathcal{C}, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet))) = \mathrm{Hom}_{D(\mathcal{O})}(L, M)$.

Proof. We have

$$\begin{aligned} H^0(\Gamma(U, \mathcal{H}om^\bullet(\mathcal{L}^\bullet, \mathcal{I}^\bullet))) &= \mathrm{Hom}_{K(\mathcal{O}_U)}(\mathcal{L}^\bullet|_U, \mathcal{I}^\bullet|_U) \\ &= \mathrm{Hom}_{D(\mathcal{O}_U)}(L|_U, M|_U) \end{aligned}$$

The first equality is (0A5X). The second equality is true because $\mathcal{I}^\bullet|_U$ is K-injective by Lemma 08BS. The proof of the last equation is similar except that it uses (0A8Z). □